Memory-Based Object Search and Route Reuse for a LiDAR Mobile Robot

ROB-UY-2004 Final Project Proposal

Abstract—This project proposes a memory-based object search system for a four-wheel mobile robot equipped with LiDAR. In a new indoor environment, the robot first explores, builds a 2D map, and searches for a designated target object. After the object is found, the robot stores the target's last known pose and a route to reach it. In later searches in the same environment, the robot first follows the memorized route while checking for the target along the way. If the target is no longer at the previous location, the robot switches back to search mode. This extends the course work on locomotion by adding map reuse, persistent memory, and adaptive research across repeated trials.

I. INTRODUCTION

Mobile robots should not only navigate safely, but also search for and repeatedly relocate useful objects. The proposed project studies a practical version of this problem: a robot must find a designated item in an indoor environment, remember where it was previously found, and use that memory to speed up future searches. The central hypothesis is that repeated searches in the same environment should become faster when the robot reuses prior spatial experience instead of searching from scratch every time.

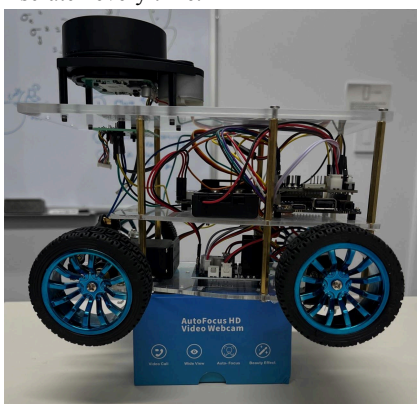

Fig. 1. Proposed LiDAR mobile robot platform.

Project Overview:

- First run: explore the environment and build a LiDAR map.
- Store the target's last known pose and a route to reach it.
- Later runs: follow the memorized route while checking for the target.
- If the target moved, fall back to search mode and update memory.

II. TECHNICAL APPROACH

The proposed system will integrate LiDAR-based SLAM, autonomous exploration, and camera-based target detection into a unified search pipeline. SLAM Toolbox will be used to perform real-time 2D mapping and localization from LiDAR data as the robot moves through an initially unknown indoor environment. To enable autonomous roaming, an exploration module will generate search goals in unexplored regions of the map, while Nav2 will plan and execute safe paths to those goals using the current occupancy map and robot pose. In parallel, a forward-facing camera will provide image data for an OpenCV-based object detection module, which will continuously check whether the designated target object is present in view. When the target is detected, the supervisory controller will interrupt exploration behavior and switch the robot into a target-approach mode. In this mode, the robot will use vision feedback to orient toward and move toward the object until it reaches a stopping distance. If the object is never detected after the room has been sufficiently explored, the robot will terminate the search and report failure.

III. PERFORMANCE METRICS

Performance will be measured over repeated trials in the same environment. Let T_i be the time to find the target in trial i, L_i the traveled path length, and C_i the number of collisions. The primary metrics are average search time, average path length, and success rate over N trials:

$$\bar{T} = \frac{1}{N}\sum T_i \#(1)$$

$$\bar{L} = \frac{1}{N}\sum L_i \#(2)$$

$$S = \frac{Number\ of\ Successful\ Trials}{N} \#(3)$$

A key comparison will be between the first search in a new environment and later searches after memory has been built. The main expected improvement is lower average time and shorter path length in repeated searches. A secondary metric is the fallback rate, defined as the fraction of repeated trials in which the remembered target location is wrong and the robot must resume exploration.

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IV. EXPERIMENT DESCRIPTION

The proposed system will integrate LiDAR-based SLAM, autonomous exploration, vision-based object detection, and memory reuse into a unified indoor search pipeline. A wheeled mobile robot equipped with a 2D LiDAR sensor and forward-facing camera will begin each trial from a known starting pose and operate in an initially unknown room. The system will use **SLAM Toolbox** for mapping and localization, **Nav2** for navigation, and **OpenCV** for target object detection.

SLAM Toolbox will perform real-time 2D mapping and localization from LiDAR data, building an occupancy grid map while estimating the robot's pose. This map will be used for navigation, exploration, and storage of target object locations. To enable autonomous roaming, the robot will use **Nav2** together with an exploration layer that generates goals in unexplored regions of the map. During an initial search, the robot will map the room while actively traversing it, with Nav2 planning collision-free paths to exploration goals.

A camera-based perception module implemented with **OpenCV** will process live image data and determine whether the designated target object is present in view. Depending on the final object choice, detection may use color thresholding, contour analysis, template matching, or feature-based matching. When the target is detected, the robot will switch from exploration mode to **target-approach mode**, cancel its exploration goal, move toward the object, and stop once it reaches it. The system will then record the object's last known location in the map.

For repeated searches in a previously mapped environment, the robot will use a **memory-guided mode**. It will first navigate to the object's stored last known location while checking for the target with the camera. If the object is found there, the trial ends successfully. If not, the robot will fall back to exploration mode and continue searching while updating the stored memory if a new location is found.

A **supervisory logic layer** will coordinate these modules by managing transitions between exploration, target-approach, memory-guided, and fallback search modes. Overall, this implementation is intended to create a mobile robot system that can search efficiently in a new

indoor environment and improve performance in repeated trials by reusing previously learned object-location information.

For the visualization of the control logic, a system block diagram is included below.

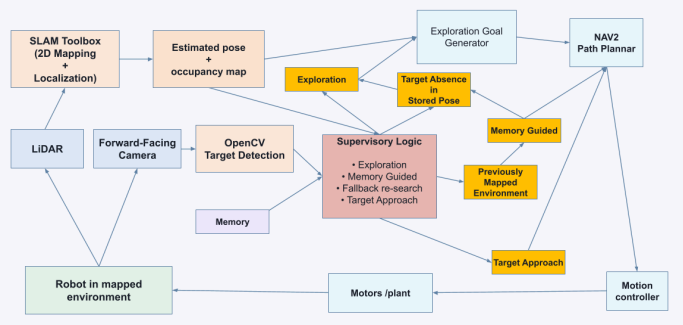

Fig. 2. Proposed control system block diagram.

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